OFFICIAL TECHNICAL DOCUMENTARY & DOSSIER

AgroScan Sentinel

An Autonomous AI & IoT Precision Agriculture Platform for Global Crop Health

1. Origin Story & Global Problem Statement

Plant diseases and insect pests destroy **20% to 40% of global crop yields** every year, causing over **$220 billion** in annual economic losses. Traditional visual inspection is slow, inaccessible to smallholder farmers, and often leads to the excessive and indiscriminate use of toxic chemicals.

AgroScan Sentinel bridges this gap by deploying edge computer vision, multimodal generative AI, smart IoT optical sticky traps, and satellite geolocation to deliver instant laboratory-grade foliar diagnosis and prescriptive treatment plans directly to farmers.

2. Core Architectural Pillars

Pillar 1: Dual AI Neural Vision Scanner

Classifies 38+ foliar phytopathogens (Late Blight, Asian Rust, Powdery Mildew) and entomological insect damage (Whiteflies, Armyworms, Aphids, Spider Mites) with 4K UHD spatial bounding box segmentation.

Pillar 2: Integrated Pest Management (IPM) & Optical Sticky Traps

Node 4B IoT optical trap integration counts insect populations before visible foliar damage occurs, alerting operators when Economic Injury Levels (EIL) are breached.

Pillar 3: Live GPS Geospatial Field Sentinel

Utilizes HTML5 browser GPS streaming with OpenStreetMap Nominatim reverse geocoding. Features an anchored Leaflet map (Street, Satellite, NDVI) and instant Autonomous Geographic Multispectral Area Scans.

Pillar 4: IoT Microclimate Telemetry & Extension Network

Live sensors monitor soil moisture (42%), Nitrogen (28 mg/kg), and soil pH (6.5), while connecting farmers with certified extension agronomists.

3. Technology Stack Specification

Layer	Technologies	Key Responsibility
Frontend UI	React 18, Vite 5, Tailwind CSS, Lucide	High-speed SPA with Daylight / Dark Sentinel theme engine
Mapping Engine	Leaflet 1.9.4, OpenStreetMap, ESRI Satellite	Interactive GPS field mapping with anchored geographic marker
Backend API	Node.js Express 5.2, CORS, REST	Report generation, auth sessions, and file streaming
AI Models	EfficientNet-B4, YOLOv8-Seg, Gemini 1.5 Flash	Dual foliar disease & pest classification, lesion bounding boxes
Database	JSON Document DB + MongoDB Mongoose	Fast local caching with automatic cloud sync failover

4. Prescriptive Treatment Philosophy

The platform delivers dual-action prescriptions:

- **Organic / Biological Bio-Controls:** Cold-pressed Neem Oil (Azadirachtin 10,000 ppm), *Bacillus thuringiensis* (Bt), *Beauveria bassiana*, and *Encarsia formosa* bio-predator wasps.
- **Targeted Chemical Treatments:** Precise dosages of *Chlorantraniliprole 18.5% SC*, *Ridomil Gold*, and *Imidacloprid* with strict pre-harvest intervals.